

Unit ID: 6798bb13aaf593b2823f122b

Unit 3 - Designing Solutions

PDF Documents

PDF Document 1: 1738062609868-Chapter 5 - unit 3 - Designing Solutions.pdf

This PDF document is included in the unit materials.

Update #1 - 2025-05-16 15:17:36



UNIT 3 DESIGNING SOLUTIONS

● Programming Concepts

WHILE STATEMENT

✓ **WHILE**



condition

While the timer is running,



task

Keep brushing your teeth.

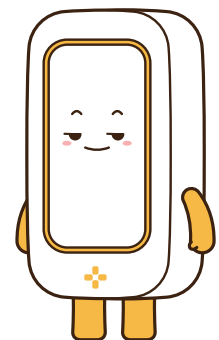
Imagine you're brushing your teeth with a timer set to make sure you brush long enough. A while loop in coding is like saying, "**Keep brushing your teeth while the timer is still running!**"

You continue brushing, and each moment, you check if the timer is still going. As long as the timer hasn't run out, you keep brushing. But once the timer stops, you stop brushing too. That's how a while loop works—it keeps doing an action (like brushing your teeth) as long as a certain condition (the timer running) is true.

TIPS

While loop keeps repeating an action as long as a condition is satisfied. In real life, a while loop can be compared to tasks such as:

"**Keep doing assignments while there are assignments left in your 'to-do' list.**"





Example

- 1 We are spooking a tired MODI, when we are near by the MODI, it's surprised. Find the condition and the action(task) in the code.

The code blocks are as follows:

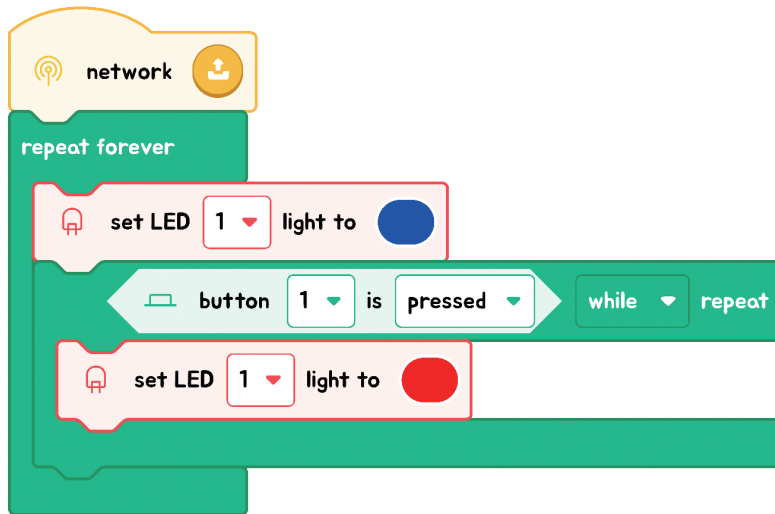
- network** (orange block with a Wi-Fi icon and a download icon)
- repeat forever** (green loop block)
 - show drawing** (pink block) with dropdowns: **Tired**, **on display**, and value **1**
 - while repeat** (green block) with dropdowns: **while** and **repeat**
 - ToF** (green block) with dropdowns: **ToF**, **distance**, and value **1**
 - <** (green block) with dropdowns: **<** and value **15**
 - cm** (green block) with dropdowns: **cm** and value **1**
 - show drawing** (pink block) with dropdowns: **Surprised**, **on display**, and value **1**

To the right of the code is a cartoon character (MODI) that is white with a red cross on its chest, red legs, and a red bell on its head. It has a surprised expression with wide eyes and an open mouth.

Below the code is a diagram for analysis:

- A light blue box labeled **Condition(s)**.
- A red box labeled **Task(s)**.
- Three arrows (blue, purple, red) pointing from the Condition(s) box to the Task(s) box.

2 Draw a flowchart based on the code shared below:



YOUR FLOWCHART



Flowcharts & Algorithms

Let's read the flowchart and create the code.

FLOWCHART

